Passenger Preferences and Perceptions of Domestic Airlines in India

Mini Project – Distribution and Inference Theory

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Tool Used: R Programming

Objective

To analyze passenger preferences for domestic airlines in India and test whether IndiGo is significantly more preferred than other airlines using statistical methods in R.

Data Summary

The dataset contained 51 valid survey responses on five factors: most travelled airline, affordability, punctuality, comfort, and customer service. Since the data was categorical, the analysis focused on the frequency of responses for each airline in every category. One response mentioning 'Nepal Airlines' was removed as it was not a domestic carrier.

Statistical Summary

To understand how evenly or unevenly respondents' choices were distributed, frequency counts of each airline were treated as numeric data. From these counts, the mean, median, variance, and standard deviation were calculated in R.

Z-Test for Proportion

A right-tailed z-test for proportion was performed to test whether the true proportion of passengers who prefer IndiGo is greater than 33.33%. The test confirmed that IndiGo's preference proportion (66.67%) is significantly higher than the expected equal-share proportion, with a p-value < 0.05. This means that the result is statistically significant, and IndiGo is genuinely more preferred among domestic passengers.

Conclusion

Through descriptive statistics and inferential testing in R, the project concludes that IndiGo is the most preferred domestic airline among respondents, while Akasa is seen as the most affordable. The use of frequency-based descriptive measures helped quantify how opinions were distributed, and the z-test statistically verified IndiGo's dominance with strong significance ($p < 0.05$).